

# Chemotherapy for chronic lymphocytic leukaemia (CLL)

Chemotherapy uses anti-cancer (cytotoxic) drugs to destroy or damage leukaemia cells. Chemotherapy for chronic lymphocytic leukaemia (CLL) is often given with targeted therapy.

## On this page

- [What is chemotherapy?](#)
- [About chemotherapy for chronic lymphocytic leukaemia \(CLL\)](#)
- [Side effects of chemotherapy for chronic lymphocytic leukaemia \(CLL\)](#)
- [About our information](#)
- [How we can help](#)

## What is chemotherapy?

Chemotherapy uses anti-cancer (cytotoxic) drugs to destroy or damage leukaemia cells. These drugs affect with the way [leukaemia cells](#) grow and divide.

## About chemotherapy for chronic lymphocytic leukaemia (CLL)

You can have the chemotherapy drugs as tablets (oral chemotherapy) or into a vein (intravenously). The chemotherapy moves around the body through the blood. It can reach leukaemia cells all over the body.

You may have either a single chemotherapy drug or a combination of different drugs given together. Chemotherapy is usually given with a targeted [therapy](#). This is called chemo-immunotherapy.

You usually have chemotherapy as several cycles of treatment. Each treatment is followed by a rest period. This lets your body recover from any side effects. The treatment and the rest period together make up a cycle of treatment.

The chemotherapy drugs most often used to treat [CLL](#) are:

- [fludarabine](#)
- [cyclophosphamide](#)
- [chlorambucil](#)
- [bendamustine](#).

Fludarabine and cyclophosphamide are usually given together with a targeted therapy called [rituximab](#). This combination is called FCR or RFC. The chemotherapy drugs can be given into a vein or as tablets.

Chlorambucil is given as tablets. It is often given along with a targeted therapy.

Bendamustine is given into a vein. It is usually given with rituximab. This treatment is called BR.

There are lots of different drugs available to treat CLL. Your doctors may use other drugs and combinations of drugs. They will tell you what treatment option they think is best for your situation.

You can usually have chemotherapy for CLL as an outpatient.

## Side effects of chemotherapy for chronic lymphocytic leukaemia (CLL)

If you are taking a single chemotherapy drug, such as [chlorambucil](#), any side effects you have will usually be mild. Treatment with a combination of 2 or more chemotherapy drugs may cause more side effects.

Side effects can include:

- risk of [infection](#)
- anaemia (reduced number of red blood cells)
- increased bruising and bleeding
- [feeling sick](#) (nausea)
- [sore mouth](#)
- [tiredness](#)
- problems with fertility ([getting pregnant](#) or [making someone pregnant](#)).

We have listed some of the most common side effects of chemotherapy. Different [chemotherapy drugs](#) have different side effects. We have more information on side effects of other drugs in our [A-Z of treatment](#).

You can talk to your doctor or nurse about what to expect from the treatment that is planned for you. We have [more information about the side effects of chemotherapy](#).

## About our information

### References

#### References



Below is a sample of the sources used in our chronic lymphocytic leukaemia (CLL) information. If you would like more information about the sources we use, please contact us at [cancerinformationteam@macmillan.org.uk](mailto:cancerinformationteam@macmillan.org.uk)

Schuh A et al. Guideline for the treatment of chronic lymphocytic leukaemia. British Society for Haematology Guidelines. July 2018.

Eichhorst B et al. Chronic lymphocytic leukaemia: ESMO Clinical Practice Guidelines for diagnosis, treatment and follow-up. ESMO Guidelines Committee. October 2020.

## Reviewers

### Reviewers



This information has been written, revised and edited by Macmillan Cancer Support's Cancer Information Development team. It has been reviewed by expert medical and health professionals and people living with cancer. It has been approved by Senior Medical Editor, [Dr Anne Parker](#), Consultant Haematologist.

Our cancer information has been awarded the [PIF TICK](#). Created by the Patient Information Forum, this quality mark shows we meet PIF's 10 criteria for trustworthy health information.

## The language we use

We want everyone affected by cancer to feel our information is written for them.

We want our information to be as clear as possible. To do this, we try to:

- use plain English
- explain medical words
- use short sentences
- use illustrations to explain text
- structure the information clearly
- make sure important points are clear.

We use gender-inclusive language and talk to our readers as 'you' so that everyone feels included. Where clinically necessary we use the terms 'men' and 'women' or 'male' and 'female'. For example, we do so when talking about parts of the body or mentioning statistics or research about who is affected.

You can read more about how we produce our information [here](#).

## Date reviewed

Reviewed: 01 January 2022 | Next review: 01 January 2025

⚠ This content is currently being reviewed. New information will be coming soon.

Our cancer information meets the [PIF TICK](#) quality mark.

This means it is easy to use, up-to-date and based on the latest evidence. Learn more about [how we produce our information](#).

## How we can help

### Macmillan Support Line

The Macmillan Support Line is a free and confidential service for people living with and affected by cancer. If you need to talk, we'll listen.



#### Call us free on 0808 808 0000

8am to 8pm, 7 days a week. You can talk to our cancer specialists about anything.



#### Email the Macmillan Support Line

Email the Macmillan Support Team if you have any questions.

#### Chat online

Available 8am to 7.45pm, 7 days a week. For confidential, real-time support if you don't want to call.



#### Online Community

Chat online anonymously to others who understand what you are going through. Our community is available 24/7 and has dedicated forums where you can get advice and ask our experts.



#### Find cancer support services in your area

What's going on near you? Find out about support groups, where to get information and how to get involved with Macmillan where you live.

**Get help**